A REVISED PARADIGM FOR ESTIMATING GARMENT COMFORT LIMITS

PROLOUGE

In a week's time, I received two enquiries from friends plus saw a post on the BPL forum from a Reddit poster all asking about specific garment's comfort limits. That was the genesis for this document. It attempts to rely on a few of my custom tables and charts, combined with minimal words, to address this topic.

One well known quote of Albert Einstein's is: "If you can't explain something simply, you don't know enough about it." That was part of the problem when I first discussed this topic in a 2007 forum thread here. After 11 years, I now know a little more about the subject. This revised document covers new insulations, new garment types, and detailed instructions so you can estimate garment comfort limits yourself.

> I believe in evidence. I believe in observation, measurement, and reasoning, confirmed by independent observers. I'll believe anything, no matter how wild and ridiculous, if there is evidence for it. The wilder and more ridiculous something is, however, the firmer and more solid the evidence will have to be. Isaac Asimov, in T*he roving mind (1April1983), p. 43*

INTRODUCTION

I think that garment knowledge is one of the UL backpacker's important assets. It doesn't cost anything, it doesn't weigh anything, and it better allows the active individual to be more comfortable in in a wide range of environments.

I measure a garment's intrinsic insulation value (Iclo) without the addition of a boundary air layer.

I enter this Iclo value into my models along with the garment type. The models look at the garment type field entry to determine what to multiply the Iclo value by. For example, a "Vest" type would be multiplied by 36%, a "Jacket" type by 48%, and a "Hoody" type by 52%. This gives me the garment's ~Iclo versus the insulation's Iclo I started with.

The models assume that the tested garment's Iclo is layered over a backpacking ensemble that totals 1 clo without any air layer.

It then adds in the appropriate air layer and calculates the thermo-neutral temperature. At thermo-neutral you are not gaining or loosing heat. In the outdoors you rarely have a perfectly still environment with a boundary layer of .75 clo like indoors. Even if you are in camp and sheltered from the wind, the

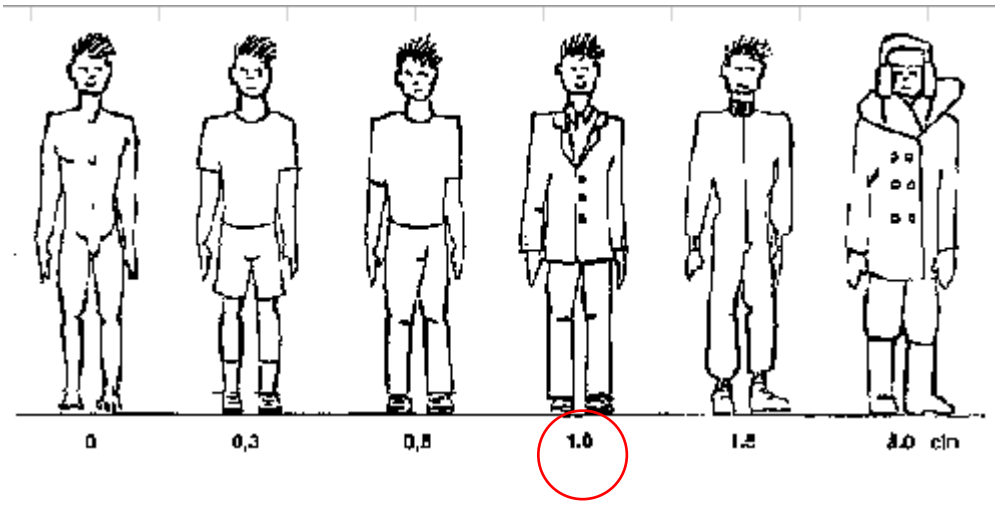
boundary layer is closer to .6 clo. When you are backpacking in the mountains the boundary layer drops to as low as .1 clo.

THREE EXAMPLES OF 1 CLO – AIR LAYER ENSEMBLES

The following two graphics illustrate typical city wear ensembles that are approximately equal to 1 clo.



Source: Innova



Source: nzdl.org

Please reference [Wikipedia-clothing insulation] for a good explanation of various street clothing items that can be combined to provide an approximately 1 Iclo insulation ensemble. Next study Figure 2, in this document, and note that sitting in your sealed living room with minimal air movement, the boundary air layer would be approximately .75 Iclo (best possible case). Add these two values together

for a total of 1.75 clo. Reference Figure P1, in this document, and look at the 1 MET line (sitting) to find the thermo-neutral temperature for 1.75 clo; it is 70F.

According to the ASHRAE-55 2010 standard, there are three methods for estimating clothing insulation using the tables that are duplicated in the afore mentioned Wikipedia article. One of which is possible to define a complete clothing ensemble as a combination of individual garments like those found in Wikipedia's Table 2. I have done this to compare the individual garments Iclo values in the Innova example ensemble with a sweater and one weight efficient UL backpacking ensemble based on the Patagonia Cap 4 hoody.

Item	Garment Iclo
Shoes	0.040
Calf-length socks	0.050
Men's briefs	0.040
Straight trousers (thick)	0.250
Dress shirt	0.250
Sweater	0.280
	0.910
Shoes	0.020
Calf-length socks	0.030
Men's briefs	0.040
Straight trousers (thin)	0.150
Cap 4 hoody	0.073
Optimally sized windshirt	0.150
Air under partial zipped windshirt	0.500
	0.963

Table o: Clothing Iclo values

CAP 4 HOODY

This garment is popular among UL backpackers and so it will be frequently included in the Iclo components that make up their ensemble. The garment Patagonia used to call a Capilene 4 hoody, they now call a Capilene® Thermal Weight Zip-Neck Hoody. The fabric they use to construct it is called Polartec Power Dry. Polartec engineered this fabric as a cool/cold weather wicking base layer and not as an insulation layer. It is constructed with Lycra to insure a close fit and a bicomponent construction. The bottom wicks and provides a small incidental amount of insulation. The dense surface weave disperses liquid water for fast evaporation from the top surface of the fabric only. It feels warm because it is very efficient at keeping your skin dry. Its lab measured warmth is low because the surface of the fabric is dense and thin.

Its insulation value is lower than a similar weight Polartec 100. Polartec 100 was engineered to provide active insulation only and does not wick liquid moisture. It does efficiently pass vapor moisture.

2018 INSULATION ANALYSIS



Figure P1: Simplified Chart for 2018



Figure P2: Simplified Chart for 2018

KEY POINTS TO REMEMBER

- For camp chores, there is a big difference in insulation required for the full possible temperature range but, for backpacking there is only a little difference. This is because our human physiology, at high activity rates, needs to dump most of your metabolically generated heat rather than retain it to be thermo-neutral. Our bodies perform best if kept at a specific internal temperature plus or minus only about 2 degrees.
- The free 7 mm air gap layer, between your base layer and windshirt, will keep you thermo-neutral in the temperature range of ~65-25. Above that temp, you need to dump heat and so you wouldn't require a windshirt. Below that temp, you would need additional light insulation. For three-seasons backpacking, that address the temperature range of most mountainous regions.
- When you combine the air permeability of the fabrics that are part of the integrated active insulations with additional shells, you need a relatively complicated calculation to determine your system air permeability. When you add a windshirt over a base layer, the windshirt's CFM is a value that can be estimated with coffee filters or tested in in a lab. When you add a high air permeability insulation, like fleece, there is only a small decrease to the windshirts previously known CFM. This can be understood by remembering the systems combined air barriers will always be less than the air barrier in the system with the lowest air barrier.

AN EXAMPLE OF HOW TO USE THIS DOCUMENT

ASSUME: YOU OWN A PATAGONIA NANO AIR HOODY AND I WANT TO USE IT FOR BOTH CAMP CHORES AND BACKPACKING IN THE OLMYPIC NATIONAL PARK. WHAT MONTHS WOULD IT BE ADEQUATE FOR EITHER OR BOTH ACTIVITY LEVELS?

USE THE INSULATION'S CLO TO CALCULATE THE GARMENT'S ICLO

We need to know the insulation value of this garment as the first step. Rarely (never that I have seen) will a vendor or a review provide you this information directly but, we can figure it out for ourselves. The Patagonia web page for this garment says " 60-g FullRange® 100% polyester stretch insulation". We look in Table 2 to find it is .56 clo/oz. We do a Google conversion to determine 60 g/m2 = 1.77 oz/yd2. We calculate .56 * 1.77 = .99 clo. We need to subtract out the completely still inside air boundary layer insulation value, used for lab testing, because we are going to use it outdoors. We know from Figure 2 that outdoors in the wind, this layer will only be ~.1 and so .99 clo - .6 = .39 Iclo. You might recall, from Figure 7, that I lab tested this garment at .38 Iclo. Synthetic insulation degrades with compression, unlike down, and your calculated value always represents the "best possible case" preceding some number of compression cycles. The Introduction and Figure 1 explained that since this garment type is a hoody, only 52% of the body surface area is covered. The garment's Iclo is .39 * .52 = .2028.

GARMENT ICLO + BASE ENSEMBLE ICLO + BOUNDARY AIR LAYER ICLO = CLO

We add the garment's Iclo to the normal backpacking clothing ensemble assumed to be 1 clo for a total of 1.2028 clo. We then add the outside, but sheltered conditions, boundary air layer to the jacket and the ensemble not covered by the jacket, which is .6 clo, for a total of 1.8028 clo.

LOOK UP THE THERMO-NEUTRAL TEMPERATURE AT 1.75 MET FOR THE CALCULATED CLO VALUE

To find the thermo-neutral temperature for 1.8028 clo doing camp chores (1.75 MET) we look up the value in Figure P2 and it is ~53F.

DETERMINE AVERAGE MONTHLY LOWS FOR THE DESTINATION AREA OF INTEREST

We next lookup the monthly weather averages for this area.

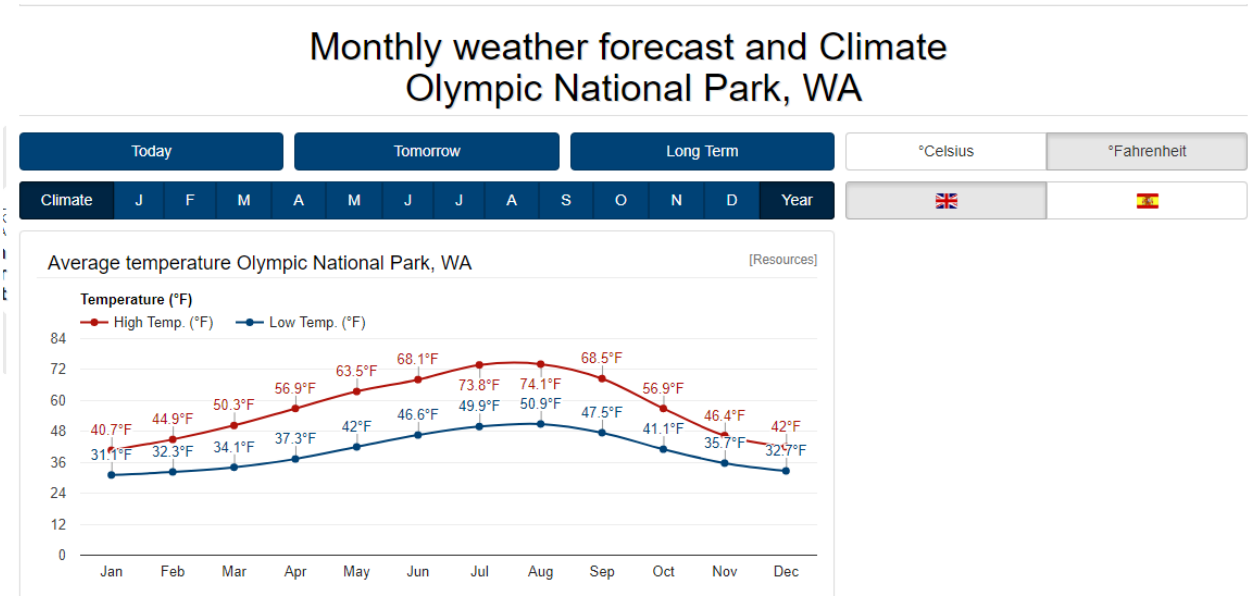


Figure 0: Monthly min/max averages

The monthly average low for every month of the year is below what is provided for this garment insulation; so, it would preclude it for consideration for camping insulation in this area.

GARMENT ICLO + BASE ENSEMBLE ICLO + BOUNDARY AIR LAYER ICLO = CLO

After determining the Iclo value for a garment, this value is input into the same model described above but, with two Iclo component adjustments. It assumes strong winds from movement, high mountain winds, and bellows losses for only an effective .1 boundary air layer. It also assumes the base ensemble is degraded 26% in warmth from accumulated sweat. These values are combined to determine the clo. The clo is then used in combination with the 7 MET line on P2 to find the thermo-neutral temperature. So, .2028 Iclo + .1 boundary layer Iclo + .74 base ensemble Iclo = .9428.

LOOK UP THE THERMO-NEUTRAL TEMPERATURE AT 7.0 MET FOR THE CALCULATED CLO VALUE

The same calculations, in combination with P2 and the 7 MET line, would next be referenced to ascertain its viable active insulation range which is 11F. Every degree above this temperature would result in progressively more sweating. With the lowest average daytime temp only being 40.7 F in January, this garment would also be precluded for use while active with the possible exception of December and January if were left largely unzipped.

YOU LOOK ONLINE AT A GARMENT YOU'RE THINKING OF PURCHASING. HOW DOES ONE DETERMINE THE CLO OF THE GARMENT?

Look in this pdf to see if it is one I already tested. If not and it is a synesthetic garment, the manufacturer should provide you the insulation type and the g/m2 of the insulation. For an uncommon synthetic insulation, not listed in my pdf Table 2, you can typically find this value by a Google search. All the remaining steps are listed in this pdf.

If it is a down garment, the manufacturer should provide you the insulation type and the oz of fill. Use this to convert a down garment's clo that I have tested and listed in Figure 7 or in the link following Figure 7. If nothing similar is found there, do a Google search "nisley & your garment name". I have done garment tests for members on BPL for about 10 years.

HOW TO ESTIMATE THE ICLO OF NEW GARMENT?

Insulation clo varies based on the density. Synthetic insulation is always manufactured at the optimal density that provides the highest clo and is, by convention, reported in clo/oz.

Down insulation continues to increase in clo for up to 2.5x the density that just fills a baffle. For that reason, down insulation information is provided in oz of fill.

Here is an example. In the pdf's Figure 7 I provided that 800 fill power (FP) down insulation Iclo for a Montbell Alpine Light Down Parka @ 5.24. Look at the Montbell specification for this garment and you will see that, in size M, it has 4.3 oz of fill. The majority of the garments used for UL backpacking have this construction and down FP. Let's say you see a down hoody you are considering from brand X that has 1.8 oz of 800 FP fill. You simply multiply 1.8/4.3 * 5.24 = 2.19.

Let's assume that another garment from brand X, you are considering, is a vest with a hood on it with 1.8 oz of 800 FP fill rather than a parka like the Montbell Alpine Light Down. The same insulation clo calculation applies to this unique garment; only the BSA calculation is unique. Per the pdf's Figure 1, we know that rather than using a body surface area of 52% for a parka, this custom garment would use a body surface area (BSA) of a vest @ 36% and a hood @ 5% BSA for a combined BSA of 41%.

Once you determine the garment's Iclo, then all of the other calculations are identical to determine the camp chores thermo-neutral temperature and backpacking thermo-neutral temperature.

METS

The MET concept represents a simple, practical, and easily understood procedure for expressing the energy cost of physical activities as a multiple of sitting metabolic rate (1 MET). 1.75 MET is defined as the average camp chores mix level, and 7 MET is the average backpacking level. The amount of insulation your body requires to be thermo-neutral is inversely proportional to your MET level. If you are below thermo-neutral, you will trend towards hypothermia. If you are above thermo-neutral, you will trend toward hyperthermia and sweating will increase as you move further from the thermo-neutral point.

CLO VALUES

1 clo is the amount of insulation that allows a sitting person (1 MET) to maintain thermal equilibrium in an environment at 70°F in a normally ventilated room.

From your backpacking wardrobe, select the underwear, shirt, pants, socks and shoes that allow you to sit in your 70 F living room comfortably and it will be ~1clo of insulation. Just like the coffee filter test to approximate CFM or the 2 cups of water in a bowl to approximate HH, you can easily approximate 1 clo for a hiking ensemble. This is the level of clothing I assume you have on, in addition to the insulation

garment that I use to calculate a camp-chores (1.75 MET) and backpacking (7 MET) thermo-neutral temperatures. My models also assume you are protected from the wind or are wearing a shell when required.

Clo values to describe insulations are by convention listed as clo oz/yd2. These values include a boundary layer of air which in the outdoors averages ~.6 clo but can be stripped away by the wind.

In addition to the "insulation clo value" to describe an insulation, there is a "garment's clo value" to describe a type of garment using that insulation. The insulation Clo value times the percent of body surface area covered is the garment's Clo value. I don't list this value but I always compute it as part of my model to determine the camp chores thermo-neutral temperature (1.75 MET) or the backpacking thermo-neutral temperature (7 MET).

DEFINITION OF TERMS USED IN SUBSEQUENT TABLES

~Equiv Fleece (active)	$	oz	mm	Hoody Equiv Iclo	Hoody Equiv Clo	CFM	~Camp Chores TN F	~Min Back pack F
Black Diamond Alpine Start Hoody	149	8.8	0.2	0.10	0.7	66	57	25

Table 1: Terms Used in Tables

$

The current suggested retail price

OZ

Size medium specification weight

MM

mm thickness of the insulation

HOODY EQUIVALENT ICLO

Whereas the boundary air layer is ~.75 clo inside your home, .6 clo is used for the camp chores rating assuming you will seek protection from the wind.

My thermal tester provides the insulation value in Iclo and my existing model to calculate the thermo-neutral temperature for camp chores and backpacking uses this value of one of the standard inputs. Hoody equivalent Iclo is the same as the Iclo except for the vest example. A vest covers 36% of the body surface area (BSA) and a jacket covers 48% of the BSA. .36/.48 = .75. So, the vest's Iclo is only 75% of an equivalent jacket.

HOODY EQUIVALENT CLO

Clo values are not required for any of my model's computations but it is for clarity when looking at synthetic insulation specifications. They provide clo per oz/yd2 which includes the boundary layer and my models add the boundary layer in to the required intrinsic value input.

~CAMP CHORES TN F

After determining the Iclo value for a garment, this value is input into a Natick Laboratories (US Armed Services Research facility) variant of the ISO TR11079 model. It computes the thermo-neutral temperature at 1.75 MET, assuming normal outdoors wind shelter, and an ensemble under the insulation layer equal to 1 clo.

~MIN BACKPACK TN F

After determining the Iclo value for a garment, this value is input into the same model described above but, with two Iclo component adjustments. It assumes strong winds from both movement and the typical mountain conditions for only a .1 boundary air layer. It also assumes the base ensemble is degraded 26% in warmth from accumulated sweat. These values are combined to determine the clo. The clo is then used in combination with the 7 MET line on P2 to find the thermo-neutral temperature.

All the garments use the same 1.75 MET and 7 MET models; only the Iclo value input is different. If an individual determines that one of their ensemble's thermo-neutral temperatures needs adjustment for their metabolism or MET rate, that same adjustment can generally be applied to all of the other garments.

BODY SURFACE AREAS

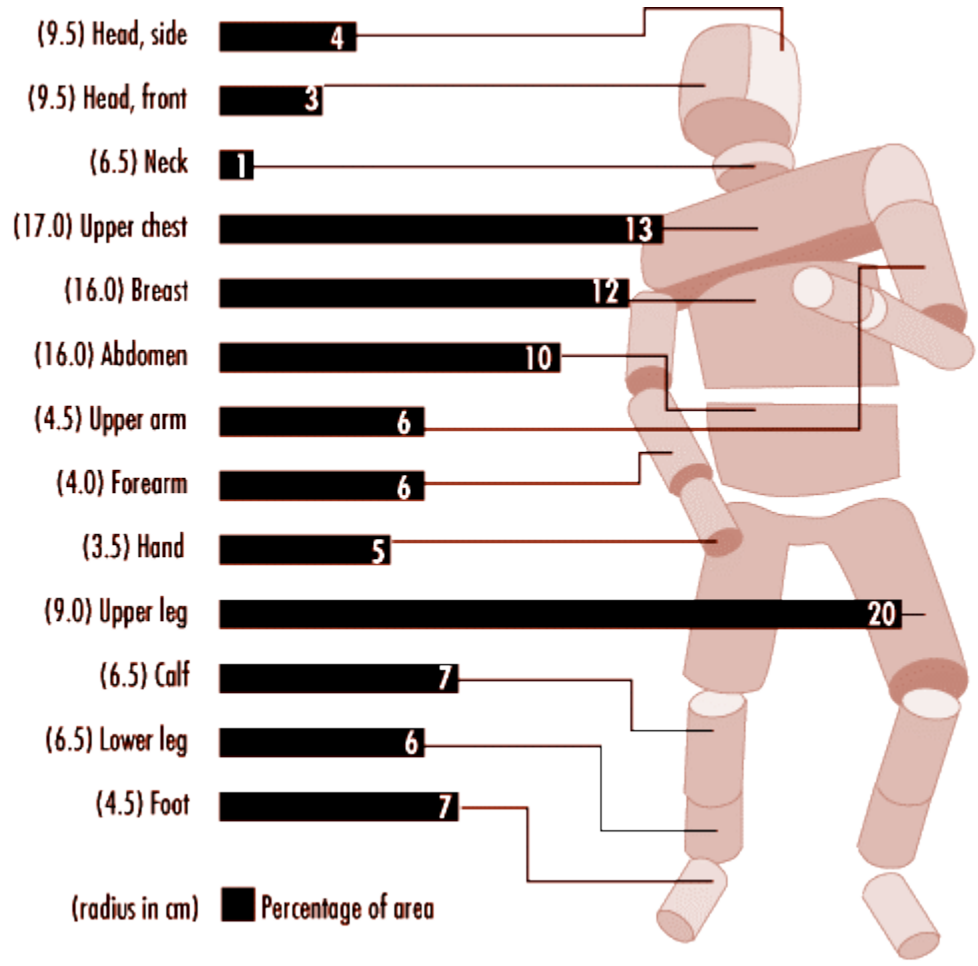


Figure 1: Body Surface Areas

BOUNDARY LAYER CLO VS WIND SPEED

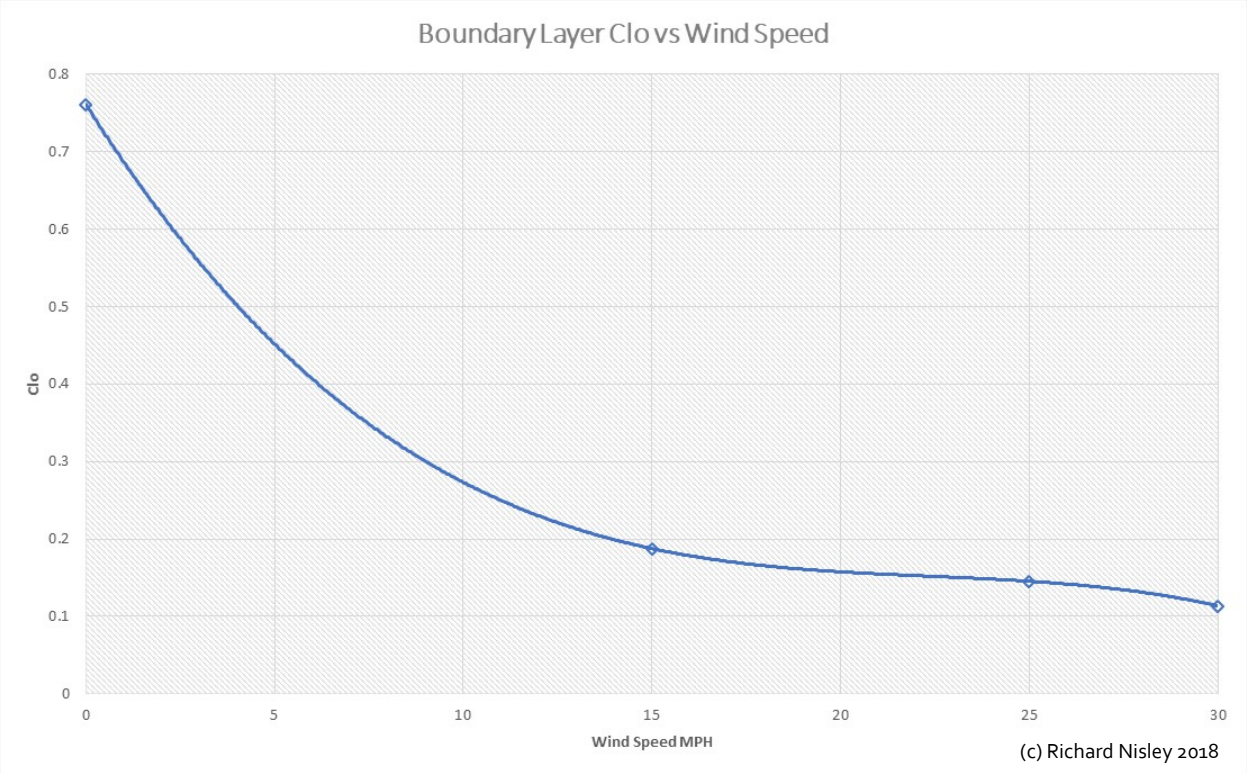


Figure 2: Boundary Layer clo Versus Wind Speed

INSULATION CLO VALUES

Insulation (Static, Active + 2 fabric layers, Active, Base)	Insul Clo oz/yd2	40/g/m2 Clo oz/yd2 **	60/g/m2 Clo oz/yd2 **	65/g/m2 Clo oz/yd2 **
900 fill Down	2.53			
800 fill Down	1.68			
650 fill Down	1.00			
Primaloft Gold aka	0.92			
Primaloft One	0.92			
550 fill down	0.85			
Primaloft Black aka	0.85			
ThermoPlume	0.85			
ThermoBall	0.85			
PlumaFill*	0.85			
Climashield APEX aka	0.82			
Climashield Apex	0.82	0.29	0.37	0.38
Coreloft	0.82	0.29	0.37	0.38
Primaloft Sport	0.79			
Primaloft Silver	0.79			
Primaloft Synergy	0.73			
Climashield Prism aka	0.76			
Thermatek	0.76			
Exceloft	0.68			
Primaloft Eco	0.68			
Primaloft Black	0.65			
Primaloft Infinity aka	0.57			
Silver Insulation Active Solution	0.57	0.20	0.25	0.27
Toray 3DeFX+ aka	0.56	0.20	0.25	0.26
Free Range	0.56	0.20	0.25	0.26
Alpha	0.38	0.13	0.17	0.18
Polartec High Loft Extreme	0.21			
Polartec High Loft Mid	0.19			
Polartec 200	0.17			
Polartec 100	0.17			
Patagonia R1	0.13			
Patagonia Cap 4 (aka Expedition Weight)	0.13			
Capiline 3	0.09			
Capiline 2	0.09			
Capiline 1	0.09			
Merino wool	0.08			
Cotton	0.04			

Table 2: Insulations clo oz/yd2

* We rank the jacket's overall warmth at a little less than the Patagonia Nano Puff, which features 60-gram PrimaLoft.

*...the Rab Xenon X Remains king of the insulated jackets. Warmer,...

** Includes 2 layers of fabric the same areal weight as the 2012 Houdini

Cold weather base layers are primarily engineered to move liquid moisture away from your skin. They are not design optimized for warmth while active or static. They are included in table 2 but should be viewed as part of your foundation ensemble rather than a dedicated insulation piece.

STATIC INSULATION

Skaha APEX Ultralight Climashield Jacket

oz	oz/yd2	clo	iclo	~Camp Chores TN F	Closest Montbell Equivalent Down Jacket	oz	iclo	~Camp Chores TN F	Closest Equivalent Synthetic Jacket	oz	Iclo	~Camp Chores TN F
									Pat Nano Puff Hoody	14.0	0.91	48
10	2.5	2.05	1.45	42	EX Light Anorak	6.1	1.84	37	MontBell Thermawrap Pro	17.6	1.42	42
									Pat Micro Puff Hooded Jacket	18.0	1.84	37
12	3.6	2.95	2.35	32	Plasma 1000 Alpine Down Parka	8.4	2.39	29	Arc'teryx Kappa Hooded Insulated Jacket	25.4	2.43	31
16	5.0	4.10	3.50	19	Mirage Parka	12.8	3.51	18	Patagonia DAS	23.6	3.26	22

© Richard Nisley 2018

Table 3: Determining static insulation temperature limits

Determine the Minimum Temperature that you anticipate for the time of year and area that you are going backpacking. Based on this minimum temperature, select the lightest Static Insulation you own that meets or exceeds that temperature.

Synthetic static insulations are only required for environments where you are not thermo-neutral and can't increase your activity level or cannot change your active layers. Examples include: mountain climbing belays, military operations, or other similarly unique situations.

Options for dealing with damp active wear after changing into dry and warm static layers, include: layer a rain shell over a windshirt to move the dew point outside the windshirt, while setting up or tearing down camp; wringing them out - swing them vigorously to remove surface water - and hang under a tarp or vestibule to further dry; or drying them inside a synthetic sleeping bag.

ACTIVE INSULATION

Please note that I know that the Black Diamond Alpine Start Hoody is not a fleece. I put it in the fleece column to illustrate that, the air gap between the base layer outer and the windshirt inner, acts as an insulation the same as would a thin fleece.

Active Insulation Garment	$	oz	Iclo	Clo	Sys CFM	~Camp Chores TN F	~Min Back pack F	~Equiv Fleece (active)	$	oz	mm	Hoody Equiv Iclo	Hoody Equiv Clo	CFM	~Camp Chores TN F	~Min Back pack F	Closest Equiv Windshirt	$	oz	CFM
Patagonia Nano Air Light Hoody	249	10.4	0.06	0.66	70.0	57	26										Black Diamond Alpine Start Hoody	149	8.8	66
								Black Diamond Alpine Start Hoody	149	8.8	0.2	0.10	0.7	66	57	25	Patagonia Airshed Pullover	119	3.7	60
Patagonia Nano Air Hoody	249	13.5	0.39	0.99	27.0	54	13										Arcteryx Squamish	159	5.5	53
								Polartec 100 (+ Equiv Beanie) exp TNF TKA or MB Chameece	55	8.8	0.3	0.47	1.07	240	53	9				
								Mountain Hardwear Monkey Man Vest (+ Beanie)	110	10.6	0.7	0.75	1.35	240	49	-2				
Arc'teryx Proton LT	299	14.4	0.97	1.57	20.5	47	-11													
								Mountain Hardwear Monkey Man (+ Beanie)	160	14.0	0.7	1.2	1.80	240	45	-20	ZSHOW Men's Super Lightweight	30	6.6	21
								Melanzana High Loft Hoodie	99	16.1	0.7	1.2	1.80	240	45	-20	Montbell UL Stretch Wind Parka	89	4.1	18

© Richard Nisley 2018

Table 4: Choosing an Active Insulation

Table 4's revised analysis 11 years later (above), is that a more typical min to max range is an effective 32F but that a base layer, optimal air gap, and windshirt is unrivaled for the entry level segment of the market now called active insulation.

Use the Minimum Temperature that you anticipate for your backpacking trip and then choose the lightest Static Insulation that meets or is higher than that temperature. It is important to note that you will be sweating in the range between the camp chore's thermo-neutral temperature and the minimum backpacking thermal neutral temperature. The more breathable your shell, the less the accumulated moisture will put you at thermal risk after slowing or stopping.

Fleece and windshirt combinations offer a wider temperature range because the shell and the insulation can be separated for use up to 70F, regardless of the fleece insulation used. This extends the applicable temperature range by an average of 20F over an integrated active insulation. Also, fleece and windshirts decrease the drying time because they can be separated to double the surface area.

Fleece also lasts much longer than non-woven synthetics. Synthetic active insulations breakdown faster than fleece or down. Although continuous insulations. like Climashield Apex, aka Arcteryx Coreloft, last longer than short staple insulations, they still appear to have rather limited lifespans. See Here.

ANDREW SKURKA'S ASSESSMENT OF ACTIVE INSULATIONS

From Here: "As a mid-layer in 3-season conditions I much prefer a fleece over a puffy. The fleece is more comfortable, breathable, and stretchy; it is warmer when wet; it has a much longer lifespan than synthetic insulations; and you can get solidly performing fleece layers for around $50.

Mid-layer puffies make sense to me in really cold conditions, when you want warmth for minimal weight. But they don't work as well as a mid-layer in 3-season conditions between a hiking shirt and shell, when you often want some kind of moisture buffer."

OPTIMAL SHELL CFM

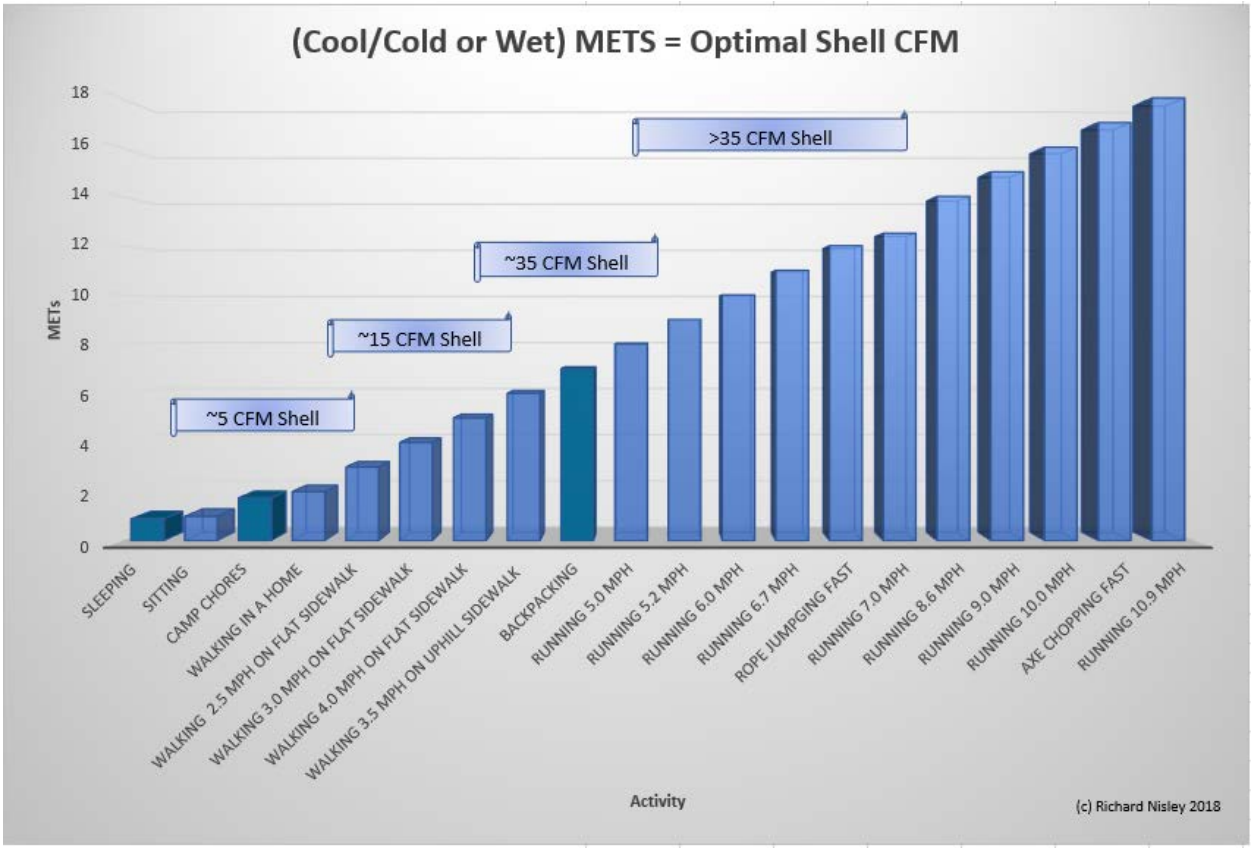


Figure 3: Optimal Shell CFM – 7 MET is the most than can be sustained daily without hourly carb intake & 18 MET the most for even minutes

7 MET IS THE MOST THAN BE SUSTAINED DAILY WITHOUT HOURLY CARB INTAKE

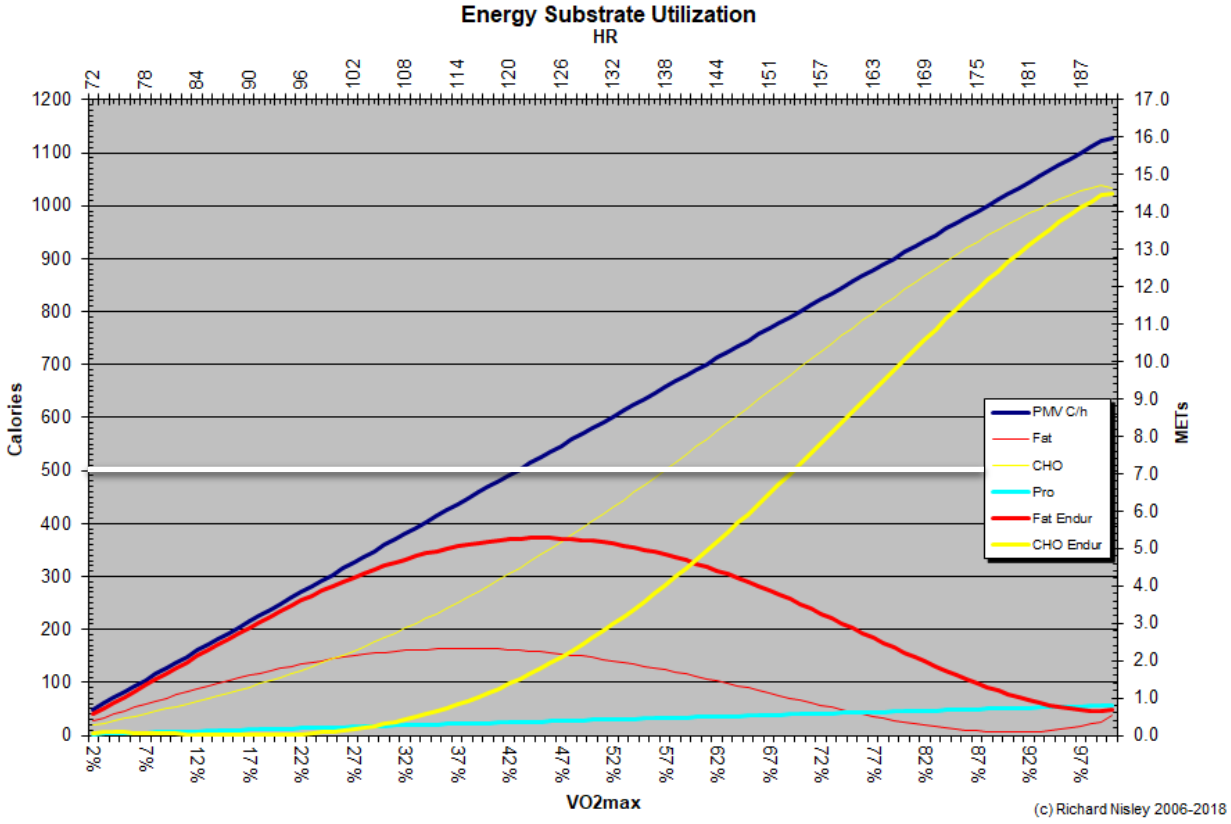


Figure 4: Light lines are initial and dark lines long term

My 2006 charts and related forum discussion are Here. Each chart is unique to the physiology of one individual. Figure 4 is one I created in 2016 for a senior citizen with only average conditioning to illustrate the fuel cross-over point for this document.

Your body burns a mix of fat and carbohydrates for its primary backpacking energy source. The highest ratio of fats to carbohydrates is at the 7 MET level. There is a cross over point from where your body burns primarily fat to primarily carbohydrate after the 7 MET level.

Assuming a 160 lb. man with 20% body fat yields 32 lbs. of fat which provides 112,000 calories. In contrast your body can only store about 1,200 calories of carbohydrates.

The food that you carry in your pack contains 9 calories for each gram of fat but only 4 calories for each gram of carbohydrate or protein. You would have to carry and then consume 35 – 75 grams of simple carbohydrates, plus their container weights, for every hour for > a ~7 MET rate.

OPTIMAL IS 35 CFM FOR 7 MET

For Here: 1/11/2013 "Patagonia constantly works with suppliers to push the limits of what is possible. We have been pushing these limits with Polartec for several decades now; the result has been many groundbreaking products over the course of those years, culminating into this pinnacle innovation. **This technology (Alpha) allows technical product designer/developers an interesting new challenge for the next several years; to 'dial in' the perfect CFM (or breathability) of a low loft garment," says Eric Neuron, Patagonia's Director of Strategic Product**. "Without the demands and feedback from Special Forces Operators, this jacket and technology would not exist. We are very proud to support their needs."

The air permeability value they selected for Level 3A (active insulation) is not available in any Patagonia or Natick document accessible via the Web. So, I acquired and tested a 2016 Patagonia 3A in multicam. The value they selected is 35 CFM for the garment (combined outer shell, Alpha insulation, and inner shell). That is both the same value as the pre-2013 Houdini and the value I have previously recommended Here as well as other forum threads for the typical 7 MET activity level of UL backpacking.

ARCTERYX PROTON LT LAYERING SYSTEM CFM

One of the problems with integrated active insulation systems is ascertaining the garment's CFM. Another problem is determining what your system's CFM is if you wear it under a windshirt. For example, assuming all the insulations average 240 CFM, if I asked you, "What is the Arcteryx Proton LT's garment's CFM?"; could you provide the answer? Spoiler… it is ~20.6 CFM which is less than the optimal 35 CFM for 7 MET activities and any CFM windshirt layered over it will just make the total CFM go even lower. Note that the 20.6 CFM value was calculated as follows: =1/((1/90)+(1/30)+(1/240)). If you assumed that the insulation had close to 0 CFM, you would come up with a value of 22.5.

Please reference the breathability section here to find 90 CFM for the exterior fabric and 30 CFM for the liner fabric.

PATAGONIA NANO AIR LIGHT LAYERING SYSTEM CFM

The Patagonia Nano Light Air, having Toray 3DeFX insulation, is specified as 70 CFM which is above the optimal CFM for 7 MET activities. What would your system's CFM be if you layered it under a 2012 Houdini? Spoiler… it is 23.3 CFM which is less than the optimal for 7 MET activities. Note that the 23.3 CFM value was calculated as follows: =1/((1/70)+(1/35)).

Please reference the breathability section here to find the 140 CFM for the outer fabric and 70 CFM for the composite total.

FLEECE + WINDSHIRT

What would a 2012 Patagonia Houdini over a fleece provide for a system CFM? Spoiler… it is 30.5 CFM, rather than the 35 CFM of just the Houdini, which is less than the optimal for 7 MET activities but much closer to the ideal than the two prior examples. This was determined from the following calculation: 1/

((1/35) + (1/240)) = 30.5. By just using a 41 CFM windshirt, similar to the Arcteryx Squamish, rather than the 35 CFM Houdini, the system CFM would be the ideal 35 CFM. This was determined from the following calculation: 1/ ((1/41) + (1/240)) = 35.0

THE COMBINED CFM REGRESSION

The key take-away is that for active insulation layering, you want the insulation component to have a very high CFM, to facilitate efficient layering.

$$AP_N = \frac{1}{\frac{1}{AP_{11}} + \frac{1}{AP_{12}} + \ldots + \frac{1}{AP_{1N}}}$$

Figure 5: How to calculate the approximate combined CFM from multiple layer's CFMs

WINDSHIRT AIRGAP INSULATION IS FREE

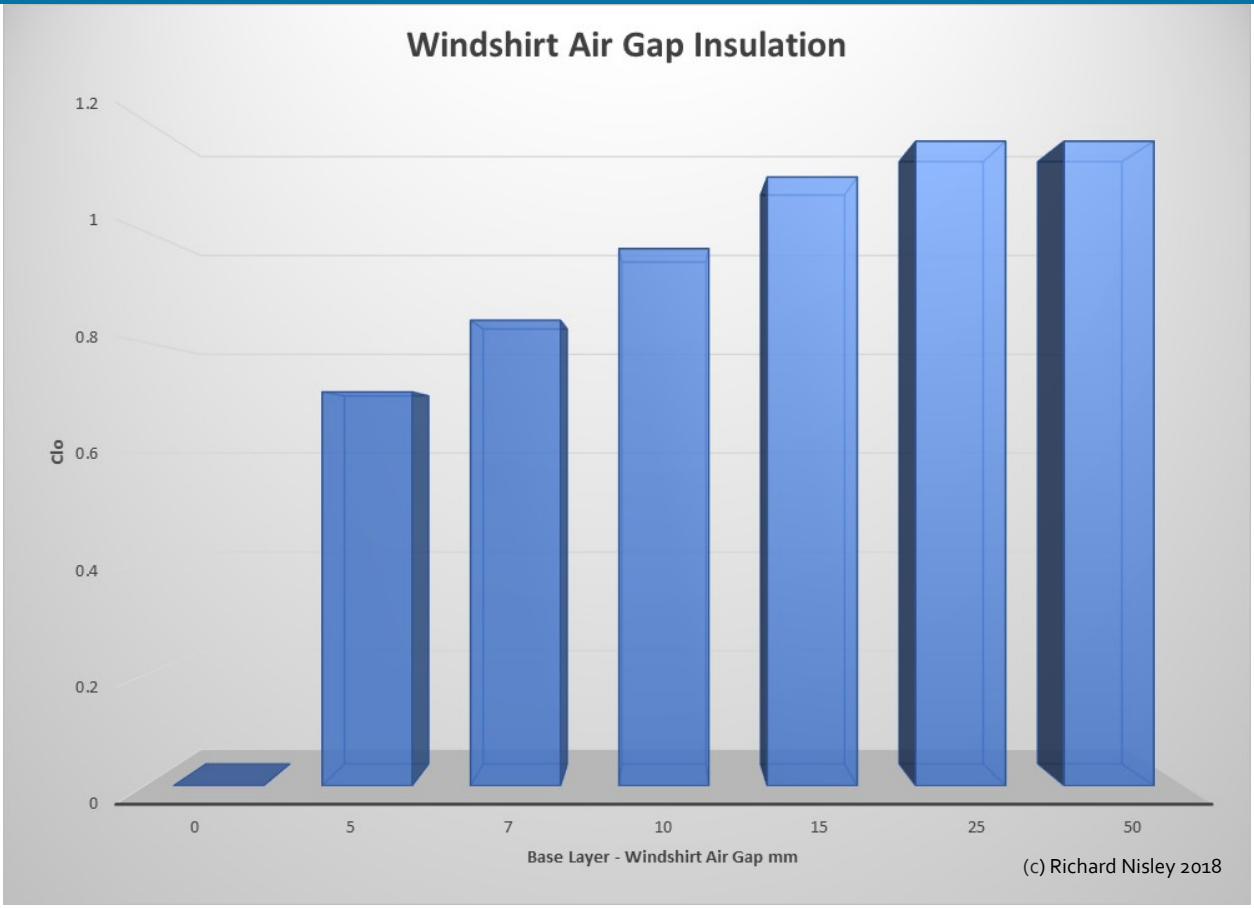


Figure 6: Windshirt Air Gap Insulation

2007

In summary, Figure 3 better illustrates the primary finding from 2007.and later years. That is why, "I created this chart to graphically demonstrate why I believe a lightweight hoody and a windshirt is the optimal clothing combination for most backpacking or hiking. For backpacking, this combination allows an effective 46F on-the-fly thermal comfort adjustment in the general range of 46F to oF."

2010

Through 2010.I was discussing the topic of the optimal windshirt air gap as "free insulation" because the now popular marketing synonym hadn't been invented yet. I don't think that most people understood… sigh.

2016

Six years later, in 2016 the old adage that, New-and-Improved sends products flying off shelves and Same Stuff- Different-Year does not, yielded a new product class they called "Active Insulation". The marketplace was excited but me…

2018

For more than a decade, the research community has published the effects of garment fit on clothing thermal insulation and moisture vapor resistance under no wind and in windy conditions. Every research study came to the same basic conclusion:

Both increase with the thickness of the air gap between the garment and the body when the air gap is small. The rate of increase gradually decreases as the air gap becomes thicker. The rate of increase is less than the theoretically ideal still air due to natural and forced convection. When the air gap is too large, thermal insulation and vapor resistance may decrease with an increasing air gap. Under no wind, the thermal insulation reaches a maximum when the air gap is about 10 mm thick, corresponding to a difference of 75 mm in girth between the garment and the body. Under windy conditions, the thermal insulation reaches a maximum when the air gap is about 6 mm thick, corresponding to a difference of 50 mm in girth between the garment and the body. More natural and forced convection is believed to cause the slower increase of thermal insulation and vapor resistance with air gap thickness in an open-structured knitted garment. The chart I created shows you approximately how much free insulation you get based on the size of the air gap between your base layer and windshirt.

In keeping with the objective of trying to keep things simpler in 2018: a Polartec 300 or Polartec Thermal Pro High Loft garment averages 7 mm thick. You should optimally size your windshirt to comfortably fit over this level of insulation. The increases beyond this point are smaller.

MY RECENT ICLO TEST VALUES USED TO PERIODICALLY TUNE MY REGRESSIONS

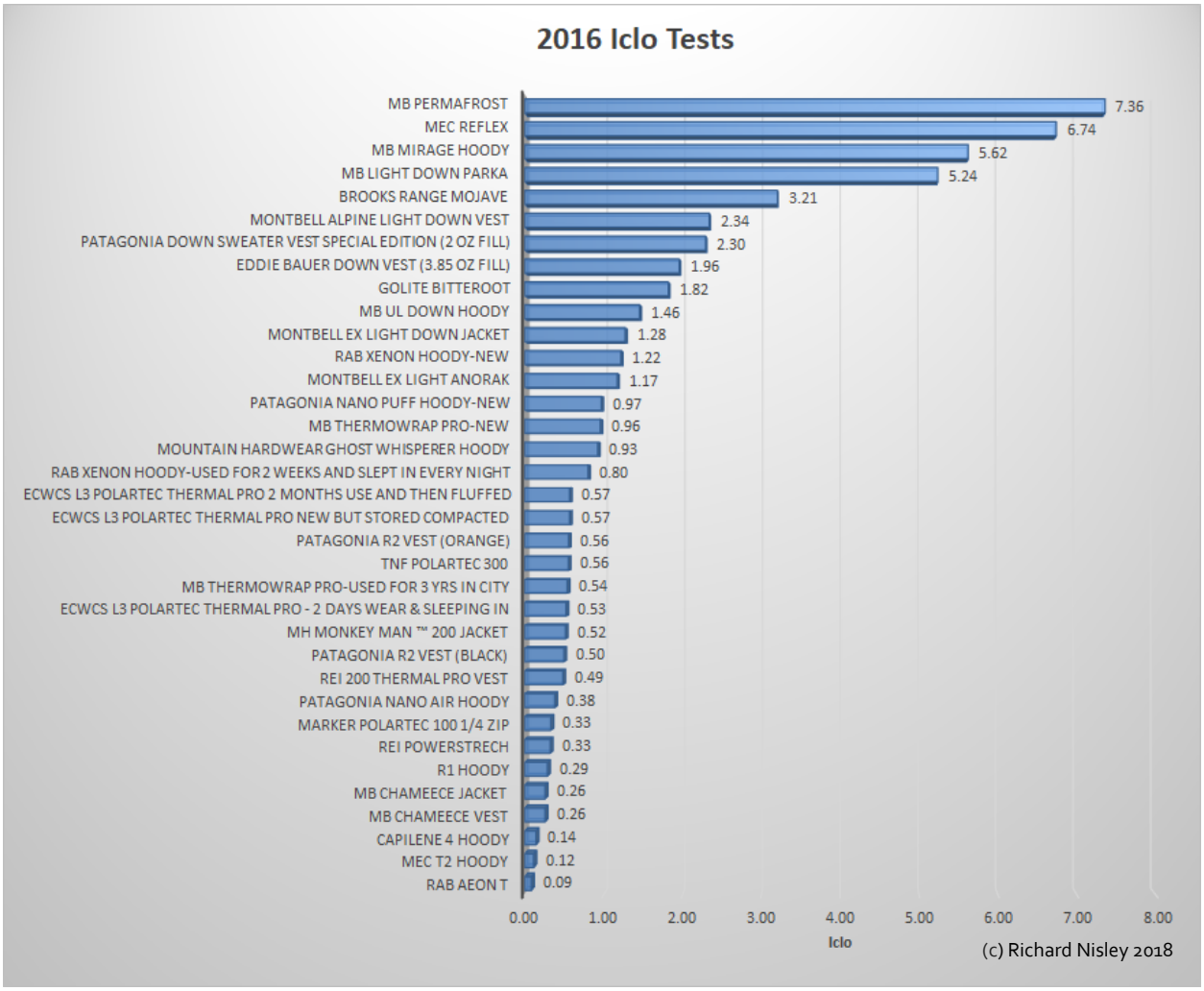


Figure 7: My 2016 Iclo tests

Some may find the [Iclo tests I published in 2009] an additional resource to understand the classification of garment categories by comparing their Iclo values.

CLOTHING INSULATION INCREASES AS THE TEMPERATURE DECREASES

The Iclo values and the clo/oz values listed in this document were measured at ~70F ambient. Most of clothing insulation's value comes from the still air in it; as the ambient temperature decreases, the insulation value increases. For three seasons use (70 -32F) there is a 6.8% increase in the insulation's value. Over the temperature range discussed in this document (70F to -20F) there is a 18.1% increase; see Figure 8 for the regression plot. This phenomenon acts as a conservative safety factor for cold weather use and is not factored out in my models. Figure 8 allows you to easily determine the insulation's increased value to factor out if desired.

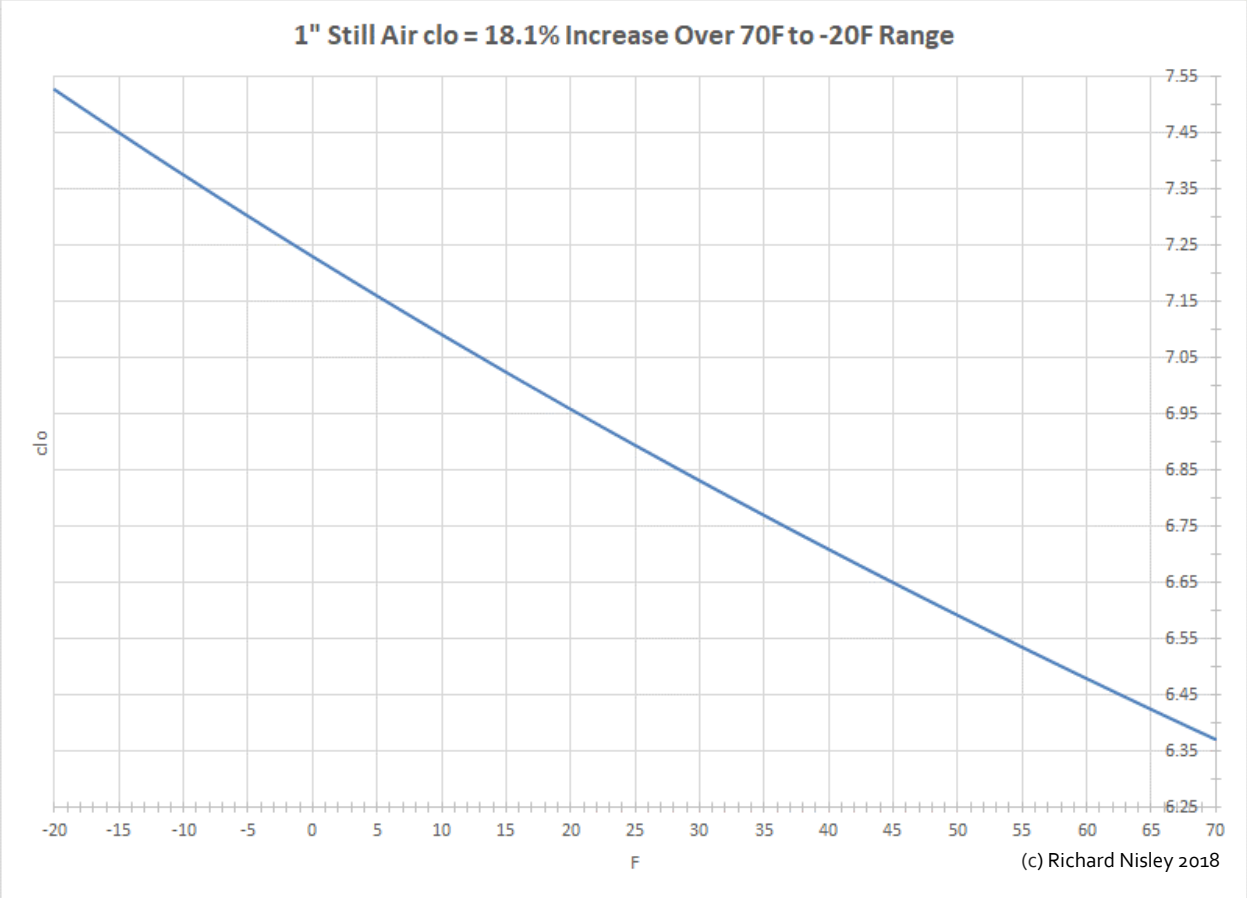


Figure 8: Insulation 1" clo value increase with decreasing F temps

IN CONCLUSION THESE POINTS BARE REPEATING

- For camp chores, there is a big difference in insulation required for the full possible temperature range but, for backpacking there is only a little difference. This is because our human physiology, at high activity rates, needs to dump most of your metabolically generated heat rather than retain it to be thermo-neutral. Our bodies perform best if kept at a specific internal temperature plus or minus only about 2 degrees.
- The free 7 mm air gap layer, between your base layer and windshirt, will keep you thermo-neutral in the temperature range of ~65-25. Above that temp, you need to dump heat and so

you wouldn't require a windshirt. Below that temp, you would need additional light insulation. For three-seasons backpacking, that address the temperature range of most mountainous regions.

- When you combine the air permeability of the fabrics that are part of the integrated active insulations with additional shells, you need a relatively complicated calculation to determine your system air permeability. When you add a windshirt over a base layer, the windshirt's CFM is a value that can be estimated with coffee filters or tested in a lab. When you add a high air permeability insulation, like fleece, there is only a small decrease to the windshirts previously known CFM. This can be understood intuitively; the systems combined air barriers will always be less than the air barrier in the system with the lowest CFM.

My thermo-neutral models only provide a reasoned starting point for determining the temperature range of various ensembles. With personal experience, you will have a more refined understanding of your actual range for each ensemble.